MODEL QUESTION PAPER 1

Public Health Engineering

Time : 3 Hour *Max.Marks : 75*

PART A

I. Answer all questions in one word or one sentence (9 x 1 = 9 Marks)

1 The rate of withdrawal of water from wells without causing failure or drying of well is known as

2 The type of joint provided to release thermal stress is called

3 The removal, deactivation or killing of pathogenic microorganisms from water is called

4 Aluminium Sulphate is used for _______

5 In separate system of sewerage, two sets of sewers are laid: one for carrying _______ and other for carrying _______

6 The minimum velocity of sewer which will prevent silting of particles is known as

7 The activated sludge contains ________ type of bacteria

8 _______ is installed to remove floating substances which include grease, oil, soap etc. from sewage

9 The traps which are used to receive waste water from bathrooms, wash basins, sinks etc. is called

PART B

II. Answer any eight questions (8 x 3 = 24 Marks)

1 Explain underground sources for water supply system

2 List various methods of forecasting population

3 List the advantages of steel pipes

4 Explain variation in rate of demand for water supply

5 Describe aeration of water

6 Differentiate conservancy system and water carriage system

7 Explain the merits and demerits of egg-shaped sewer

8 Sketch the component parts of manhole

9 Define BOD

10 Explain grit chamber

PART C

Answer ALL questions. Each question carries 7 marks (6 x 7 = 42Marks)

III Explain infiltration gallery with neat sketch

OR

IV Explain methods of conveyance of water

V Explain various methods of distribution of water

OR

VI Draw a neat sketch showing layout of rapid sand filter and explain its working

VII Illustrate the layout of "dead end system" and mention the advantages and disadvantages

OR

VIII Distinguish slow sand filters and rapid sand filters

IX Explain combined sewerage system and its suitability

OR

X Explain "Separate sewerage system" and mention the advantages.

XI Draw the sketch of flushing cistern and explain its working

OR

XII Explain the classifications of traps in house drainage system

XIII Describe advantages and disadvantages of activated sludge process

OR

XIV Explain different methods of sewage disposal

SCHEME OF VALUATION

Course Title:Public Health Engineering

PART A

I. Answer all questions in one word or one sentence (9 x 1 = 9 Marks)

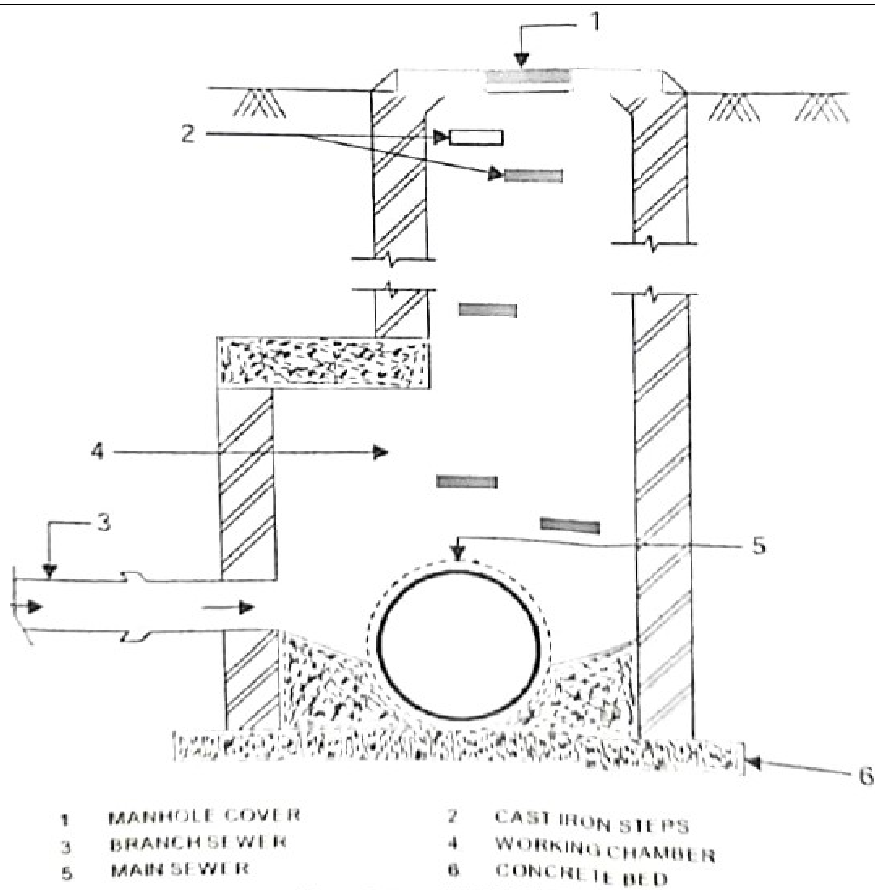
Qst No.	Scoring Indicator	Split up score	Total
1	Yield of well	1	1
2	Expansion joint	1	1
3	Disinfection	1	1
4	Coagulation	1	1
5	Sewage Storm water	0.5 0.5	1
6	Self-cleansing velocity	1	1
7	Aerobic	1	1
8	Skimming tank	1	1
9	Gully trap	1	1

PART B

II. Answer any eight questions (8 x 3 = 24 Marks)

Qst No.	Scoring Indicator	Split up score	Total
1	There are also called subsurface sources. When rain falls on the ground apart of its percolates down ward under the influence of gravity until itreaches an impervious stratum eg:- springs, infiltration galleries, infiltration wells etc.	1.5 1.5	3

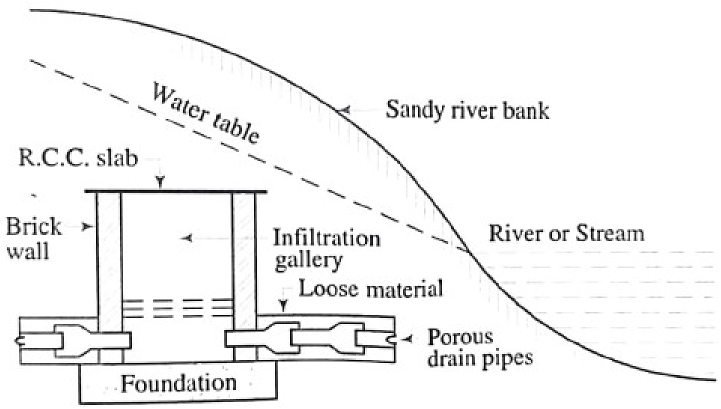
2	1.Arithmetical increase method 2.Geometrical increase method 3.Incremental increase method 4.Graphical method 5.Comparative method 6.Zoning method 7.Ratio and correlation method 8.Growth composition analysis method 9.Logistic curve method	Any 6 3	3
3	Advantages 1. Stroger and cheaper than Cast Iron pipes 2. Withstand higher internal pressure 3. Lighter and hence easy to handle and transport 4. Obtainable in large lengths, thus minimizing number of joints	Any 3 3	3
4	• It is seen that demand is not uniform throughout the year but varies widely in different seasons, in different months of the year, in different days of the month and in different hours of the day • These variations are due to many factors such as habits of people,climatic conditions, types of industries etc	3	3
5	Aeration is essentially a physical or mechanical method by which water is brought in close contact with air so as to absorb oxygen for the reduction of taste, odour etc.by virtue of oxidation and expulsion.	3	3
6	<u>Conservancy system</u> The night soil is collected in pans from dry latrines, conserved for the complete day and finally removed by scavengers to the disposal site through trucks or carts <u>Water carriage system</u> Water is used as a medium for conveying sewage through the underground sewers.	1.5 1.5	3
7	<u>Merits</u> • Self-cleansing velocity will be available even during small discharges • Suitable for combined sewerage system as well as separate system • Better hydraulic properties compared to circular sections <u>Demerits</u> • Construction is difficult • Less stable than circular sections	3	3

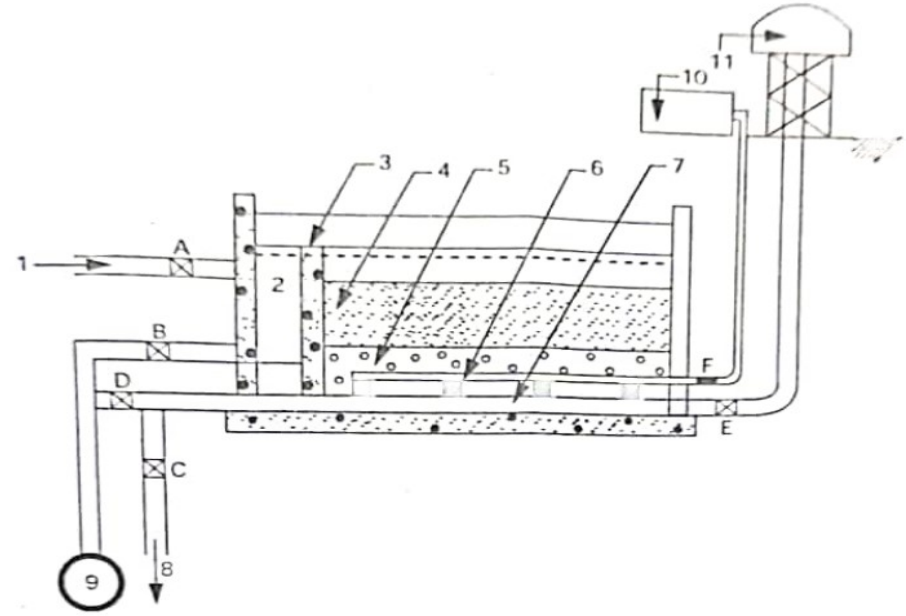
8	 MAN-HOLE	3	3
9	The amount of oxygen required for microbs to carry out the biological decomposition of dissolved solids or organic matter in sewage under aerobic conditions at standard temperature is known as B.O. D	3	3
10	• The purpose of providing grit chamber in the sewage treatment process is to remove grit, sand and such other inorganic matter from sewage. • To achieve this, the velocity of flow in grit chamber is decreased to such an extent that the heavier inorganic materials settle down at bottom of grit chamber and lighter organic materials carried forward for further treatment	3	3

PART C

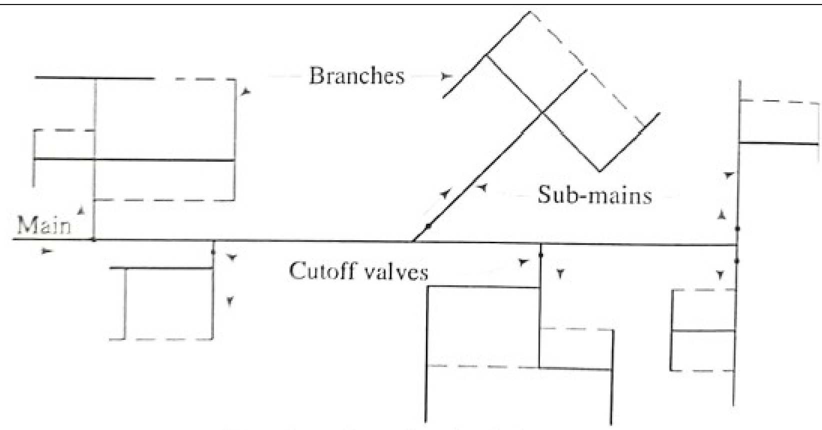
Answer ALL questions. Each question carries 7 marks (6 x 7 = 42Marks)

Qst No.	Scoring Indicator	Split up score	Total
III	• Infiltration galleries are the horizontal tunnel like wells constructed in open cut 3 to 4 m deep along the bank or in the bed of rivers • The galleries are covered with masonry arches of R C C slabs • The side walls are provided with number of weep holes. The spaces between sides of the trench and wells are filled with graded gravel and pebble stones to increase the intake capacity • Water is collected in the sump constructed at the end and pumped out	4	7

	 Infiltration gallery	3	
IV	Water has to be conveyed from the source to the treatment plant and subsequently distributed for consumption. For this purpose, theconduits are used. <u>Gravity conduits</u> Flow in the conduit due to gravity. The water surface is free i.e. subjected to atmospheric pressure Gravity flow occurs due to the falling gradient provided to theconduit Eg. canals, flumes, aqueduct <u>Pressure conduits</u> These also called pipe conduit in which large amounts of water flows under pressure They are of circular cross – section made of different materials like cast iron, steel, R.C.C etc.	1 3 3	7
V	The following are the four main methods of distribution of water 1. Dead end method or tree system 2. Grid iron method 3. Circle or ring system 4. Radial system <u>1.Dead end methodor tree system</u> • It consists of one supply main from which sub mains are taken. The submains again divide into several branch lines from which service connections are given to the consumers <u>2. Grid iron method</u> • In this system, the mains, submains and branches are interconnected with each other <u>3. Circle or ring system</u> • In this system, the entire locality is divided into either rectangular or circular blocks <u>4.Radial system</u> • This system is the reverse of ring system, water flows radially from one point to the outer periphery	1 1.5 1.5 1.5 1.5	7

VI	 1. INLET 2. INLET CHANNEL 3. OVER FLOW WEIR 4. SAND MEDIUM 5. GRAVEL BED 6. LATERAL DRAINS 7. MANIFOLD DRAIN PIPE 8. TO CLEAR WATER RESERVOIR 9. WASH WATER DRAIN 10. AIR COMPRESSER UNIT 11. OVER HEAD TANK TO SUPPLY WASH WATER A, B, C, D, E AND F ARE VALVES RAPID SAND FILTER The effluent from sedimentation with coagulation unit is applied to the filter through valve A. Valves B, D and E are closed and then water is allowed to percolate through sand bed. Water is filtered by the action of different mechanisms of filtration. The filtered water collected in the under-drainage system is taken to the clear water reservoir through manifold drain pipe by opening valve C.	4 3	7

<table>
<tr>
<td>VII</td>
<td>



Dead-end method of layout

<u>Advantages</u>

- It is relatively cheaper and simpler system. The expansion and extension are easy
- The discharges and pressures at any point in the network can be easily calculated and hence design is easy
- Requires lesser number of cut off valves
- Shorter pipe lengths and smaller diameters are needed and the laying of pipes is also easier

<u>Disadvantages</u>

- If there is any damage or repair in any pipeline, the complete area fed by that line cannot get water supply
- There are number of dead ends where the water becomes stagnant leading to accumulation of sediments and deterioration of quality
- During fire hazards, the supply cannot be increased by diverting supplies from any other side

</td>
<td>2

2.5

2.5</td>
<td>7</td>
</tr>
<tr>
<td>VIII</td>
<td>

Item	**Slow sand filter**	**Rapid sand filter**
Base material of gravel	Varies from 3 to 65 mm in size and 300 to 750 mm depth	Varies from 3 to 40 mm in size and 600 to 900 mm in depth
Coagulation	Not required	Essential
Construction	Simple	Complicated
Cost of operation	low	high
Filter media of sand	Effective size varies from 0.2 to 0.3 mm and uniformity coefficient is about 2 to 3	Effective size varies from 0.35 to 0.60 mm and uniformity coefficient is about 1.2 to 1.7
Period of cleaning	1 to 3 months	2 to 3 days
Skilled supervision	Not essential	essential

</td>
<td>7</td>
<td>7</td>
</tr>
<tr>
<td>IX</td>
<td>

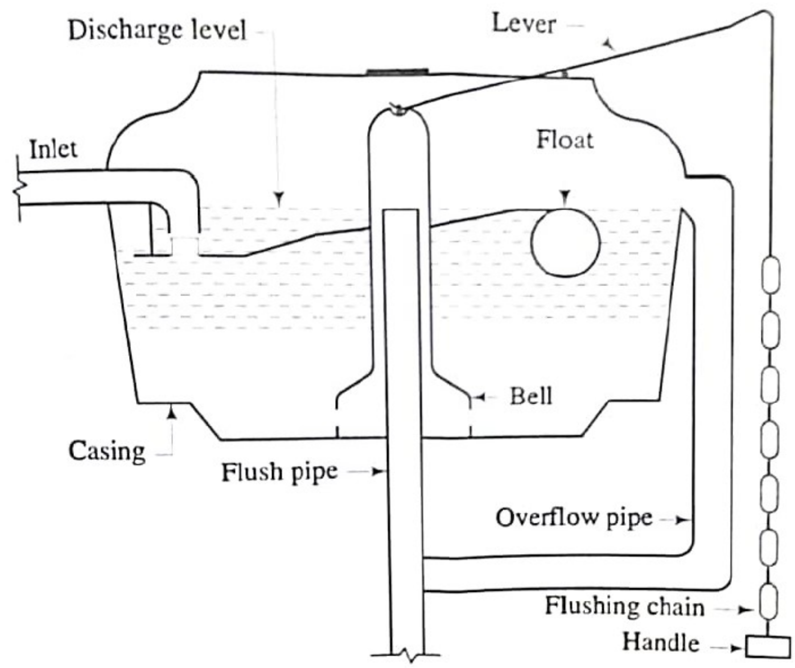
<u>Combined system</u>

In this system only one set of sewers is laid which carry the domestic and industrial sewage and also the storm water.

<u>Suitability</u>

This system is suitable for areas with small and evenly distributed rainfall. It is also suitable for situations where combined sewage has to be pumped

</td>
<td>7</td>
<td>7</td>
</tr>
</table>

	and also for heavily built-up areas with a little space for laying.		
X	<u>Separate system</u> In this system there are two separate sets of sewers One set carries domestic and industrial sewage and another set carries storm water <u>Advantages</u> • Sewers are smaller and hence economical • Quantity of sewage to be treated is smaller and hence cost of treatment less • There is no risk of pollution due to over flowing of storm water from sewers • The quantity of sewage to be pumped if necessary, is small. Hence cost of pumping is low	2.5 4.5	7
XI	• Flushing cistern consists of a bell connected to flushing chain through a lever. • When the chain is pulled, the bell is lifted up and the water in the tank rushes through the flushing pipe by syphonication. • The float valve now allows water from the inlet into the cistern and thus the cistern is made ready for next flushing  Flushing cistern	3 4	7
XII	<u>Classification of traps according to shape</u> 1. P – trap: - having the shape of letter "P". the legs are right angles to each other 2. Q – trap: - it resembles the letter Q. the legs are at some angle other than 90. 3. S – trap: - it has the shape of letter "S". the legs of traps are parallel to each other <u>Classification of traps according to use</u> 1. Floor– trap: - Used for admitting surface wash or waste water from	3.5	7

	floors of bath rooms, kitchens 2. Gully traps: - used to receive the sullage from baths, sinks, wash basins and rain water from house tops 3. Intercepting traps: - used at junction of the house drain and public sewer in order to prevent the foul gases produced in sewers from entering into the house drainage system	3.5	
XIII	Advantages • High efficiency in the removal of B.O.D and solids by about 90% • Free from fly nuisance and bad smell • Area required for construction is small • Initial cost is less • Loss of head in the process is small Disadvantages • Its operational cost is high • Skilled supervision is needed • The change in quality and quantity of effluent upset the process • Large volume of sludge is produced	3.5 3.5	7
XIV	There are mainly four methods of disposal of sewage 1. Disposal on land 2. Disposal in water 3. Direct and indirect reuse of waste water 4. Artificial methods <u>1.Disposal on land</u> • Before depositing on land, it is verified whether the effluent is treated and removed off the pollutants to such an extend to satisfy the standards for disposal <u>2. Disposal in water</u> • It is also called dilution technique • This method involves disposing sewage in natural body of water, taking the advantages of its self-purification <u>3. Direct and indirect reuse of waste water</u> • The reuse of treated water effluent by direct or indirect means <u>4.Artificial methods</u> • Oxidation ponds • Oxidation ditches • Aerated lagoons etc.	1 1.5 1.5 1.5 1.5	7

Course:**Public Health Engineering**

BluePrint

MarkDistribution

Module	hr/ Module	Marks/Module $(h_i/\sum H_i)$*123(±5%)	TypeofQuestions							
			PartA		Part B		PartC		Total	
			NoofQuestions	Marks	NoofQuestions	Marks	NoofQuestions	Marks	NoofQuestions	Marks
1	18	30	2	2	4	12	2	14	8	28
2	18	31	2	2	1	3	4	28	7	33
3	18	30	2	2	4	12	2	14	8	28
4	19	32	3	3	1	3	4	28	8	34
Total	73	123	9	9	10	30	12	84	31	123

BluePrint

CognitiveLevelMarkDistri bution

CognitiveLevel	Marks	%ofMarks
Remembering	18	15
Understanding	105	85
Applying	0	0
Analysing		
Evaluating		
Creating		
Total	123	100

Question wise analysis

Course:**Public Health Engineering**

Q.No	ModuleOutcome	CognitiveLevel	Marks	Time
I.1	M 1.01	Remembering	1	2
I.2	M 1.04	Remembering	1	2
I.3	M 2.01	Remembering	1	2
I.4	M 2.02	Remembering	1	2
I.5	M 3.02	Remembering	1	2
I.6	M 3.01	Remembering	1	2
I.7	M 4.01	Remembering	1	2
I.8	M 4.02	Remembering	1	2
I.9	M 4.04	Remembering	1	2
II.1	M 1.01	Understanding	3	8
II.2	M 1.02	Remembering	3	8
II.3	M 1.04	Remembering	3	8
II.4	M 1.02	Understanding	3	8
II.5	M 2.02	Understanding	3	8
II.6	M 3.02	Understanding	3	8
II.7	M 3.03	Understanding	3	8
II.8	M 3.03	Understanding	3	8
II.9	M 3.04	Remembering	3	8
II.10	M 4.02	Understanding	3	8
III.	M 1.01	Understanding	7	16
IV.	M 1.04	Understanding	7	16
V.	M 2.05	Understanding	7	16
VI.	M 2.03	Understanding	7	16
VII.	M 2.05	Understanding	7	16
VIII.	M 2.03	Understanding	7	16
IX.	M 3.02	Understanding	7	16
X.	M 3.02	Understanding	7	16
XI.	M 4.04	Understanding	7	16
XII.	M 4.04	Understanding	7	16
XIII.	M 4.01	Understanding	7	16
XIV.	M 4.03	Understanding	7	16

MODEL QUESTION PAPER 2

Public Health Engineering

Time : 3 Hour *Max.Marks : 75*

PART A

I. Answer all questions in one word or one sentence (9 x 1 = 9 Marks)

1 Spring is an example of _______ source of water

2 _______ is the amount of water supplied per person per day to satisfy the various demands

3 List any two purposes for which water is required

4 Zeolite process is used to remove ________ in water

5 The application of chlorine beyond the stage of break point is known as

6 The system is which the sewage and storm water are carried in a single sewer line is

7 The under-ground conduit conveying sewage is called ________

8 ________ traps are used at the junction of the house drain and public sewer in order to prevent the foul gases produced in sewers from entering into the house drainage system

9 To remove floating matters like papers, wood, cork, fibre, grass etc. the unit used is ________

PART B

II. Answer any eight questions (8 x 3 = 24 Marks)

1 List the factors affecting rate of demand of water

2 Explain geometrical increase method used for forecasting population

3 Explain specific yield of a well

4 Describe infiltration gallery

5 Differentiate pre–chlorination and post-chlorination

6 Define chemical oxygen demand

7 Differentiate between sewage and sullage

8 List the factors influencing selection of shape of sewer

9 List the merits of separate sewerage system

10 Explain the purpose of installing a skimming tank in primary treatment of sewage

PART C

Answer ALL questions. Each question carries 7 marks (6 x 7 = 42Marks)

III Explain construction of infiltration well with neat sketch

OR

IV Explain socket and spigot joint used for connecting water supply pipes

V Illustrate the layout of grid iron method and mention the advantages and disadvantages

OR

VI Draw a neat sketch showing layout of pressure filter and explain its working

VII List the requirements of a good coagulant

OR

VIII Explain break point chlorination

IX Draw the neat sketch of drop manhole and explain its construction

OR

X Explain water carriage system of sewage disposal

XI Explain the objectives of sewage treatment

OR

XII Describe different methods of sludge disposal

XIII Explain the importance of biogas plant

OR

XIV Draw neat sketch of floor trap, gully trap and intercepting trap

SCHEME OF VALUATION

Course Title:Public Health Engineering

PART A

I. Answer all questions in one word or one sentence (9 x 1 = 9 Marks)

Qst No.	Scoring Indicator	Split up score	Total
1	Subsurface sources of water/ground water sources	1	1
2	Per capita demand	1	1
3	1)Domestic demand 2)Industrial and commercial demand 3)Public demand 4)Losses and wastes 5)Fire demand	Any 2 1	1
4	Hardness of water	1	1
5	Super chlorination	1	1
6	Combined system	1	1
7	Sewer	1	1
8	Intercepting traps/interceptors	1	1
9	Screens	1	1

PART B

II. Answer any eight questions (8 x 3 = 24 Marks)

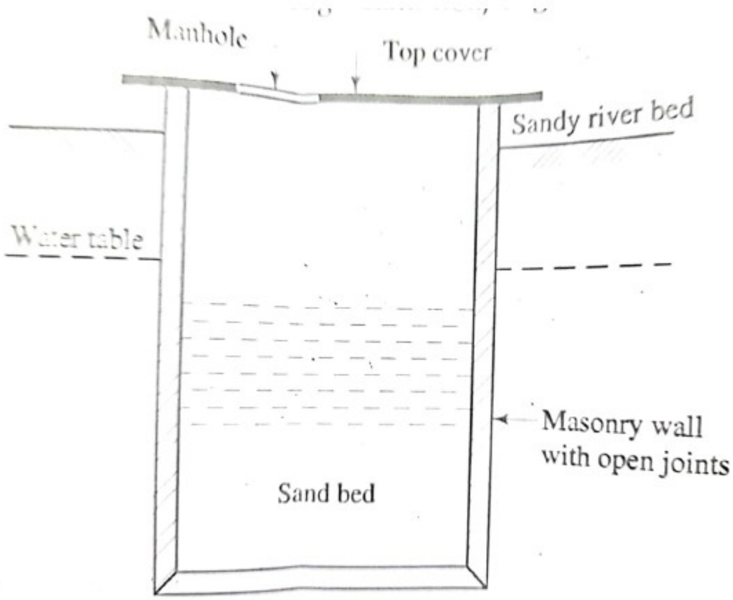
Qst No.	Scoring Indicator	Split up score	Total
1	1.Climatic conditions 2.Cost of water		

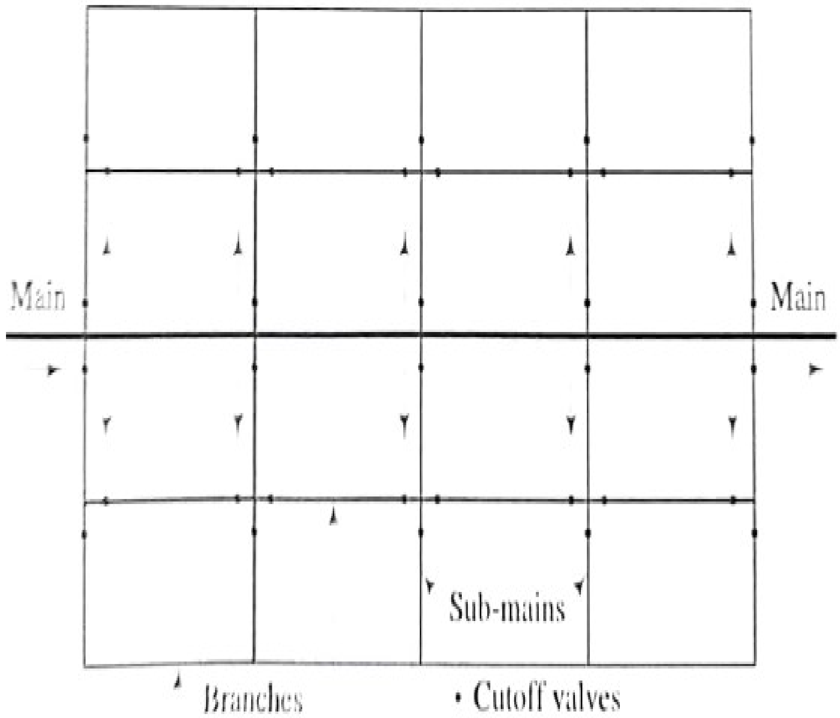
	3.Distribution pressure 4.Social and economic status of people 5.Industries and commerce 6.Quality of water 7.System of supply	Any 6 3	3
2	In this method the percentage increase in population from decade todecade is assumed to remain constant The population at the end of nth decade Pn can be estimated as $P_n = P(1+r/100)^n$ WhereP_n= Population at the end of 'n' years or decades P = Present population r = average percentage increase per year or decade	1.5 1.5	3
3	The term yield of a well is used to indicate the rate of withdrawal or pumpingof water from wells without causing failure or drying of well. Specific yield of a well is defined as the yield per unit draw down. i.eyield per one meter draw down	3	3
4	• Infiltration galleries is a subsurface source of water • The horizontal tunnel like wells constructed in open cut 3 to 4 m deep along the bank or in the bed of rivers. The galleries are covered with masonry arches of R C C slabs	1 2	3
5	**Pre-chlorination** • It is the application of chlorine before filtration. • It reduces bacterial load on filters and helps inremoving tasteand odour and makes the water fit for us **Post-chlorination:** • When chlorine is added in the water after all treatments • it is generally done after filtration. Thechlorine is commonly added in the clear water reservoir	1.5 1.5	3
6	Chemical Oxygen Demand or C.O.D is a measurement of the oxygenrequired to oxidize soluble and particulate organic matter in water.	3	3
7	• Sewage is waste water from a community,containing solid and liquid excreta, derivedfrom houses, street and yard washing, factoriesand industries. • The term sullage is applied to waste waterwhich does not contain excreta, e.g. wastewater from kitchen and bathrooms	3	3
8	1. Efficiency of flow which depends on hydraulic mean depth 2. Structural stability 3. Cost of sewer 4. Ease of construction and maintenance 5. Resistance to internal and external pressure 6. maintenance of self-cleansing velocity	Any 3 3	3
9	1. Sewers are smaller hence economical 2. Quantity of sewage to be treated is smaller and hence cost of treatment is less 3. There is no risk of pollution due to overflowing of storm water from sewers 4. The quantity of sewage to be pumped if necessary, is small. Hence cost of pumping is low	Any 3 3	3
10	A skimming tank is a chamber so arranged that the floating matter		

	likeoil, fat, grease etc., rise and remain on the surface of the waste water(Sewage) until removed, while the liquid flows out continuously underpartitions or baffles.	3	3

PART C

Answer ALL questions. Each question carries 7 marks (6 x 7 = 42Marks)

Qst No.	Scoring Indicator	Split up score	Total
III	• These are vertical shallow wells constructed inseries along the banks of rivers. The wells areconnected by porous pipes • The infiltration wells are constructed either bymasonry concreteor by sinking R C C rings • The wells are covered atop with R C C Slab Manholes are provided in the top slab forinspection purpose • Infiltration wells are connected by porous pipes tothe slump well called jack well Water collectedfinally in the jack well and is pumped, treated ifnecessary and then distributed  Infiltration well	4 3	7
IV	• Socket and spigot joint also called Bell and spigot joint • It is widely used for joining cast iron pipes • Each pipe consists of one enlarged end (socket end) and another normal end (spigot end) • The spigot end is wound round with stands of hemp yarn and is tightly fitted into the socket end • A rubber gasket or joint runner is then clamped in place around the joint so that it fits tightly around the outer edge of the bell so that hot lead will not run out of the joint space • Molten lead is then poured to fill the remaining space of the socket • After lead shrinks on cooling and is expanded using caulking tool and a hammer	7	7

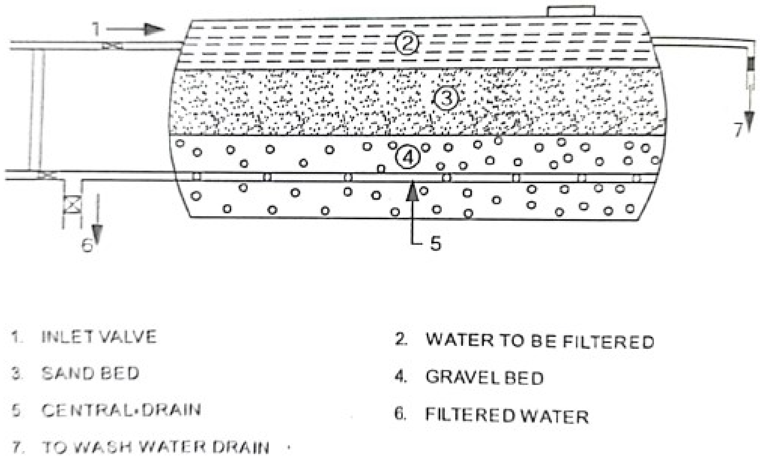
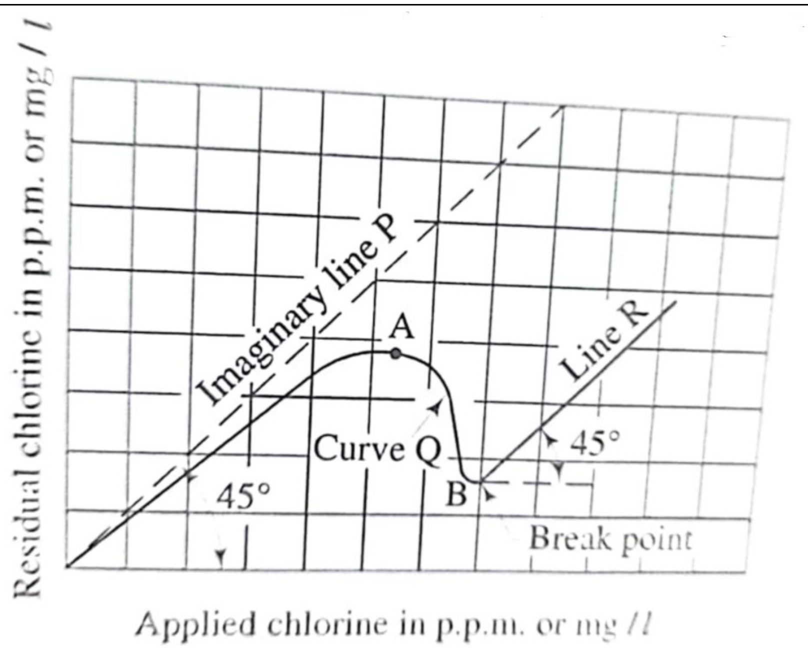
<table>
<tr><td>V</td><td>

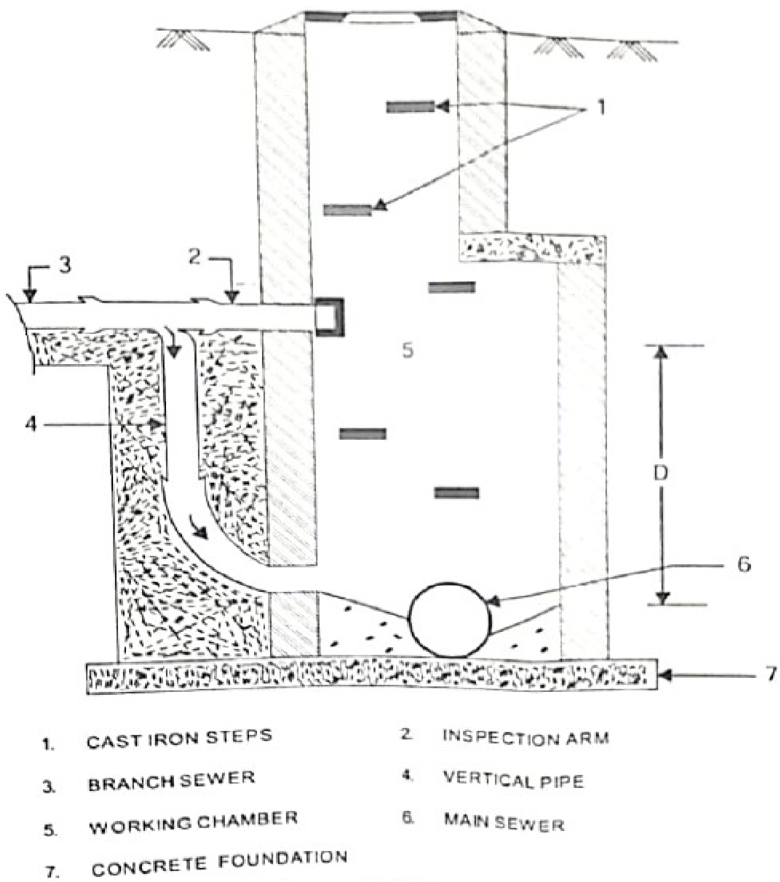
Grid-iron method of layout

<u>Advantages</u>
There are no dead ends and hence stagnation of water and its consequences are eliminated
In case of any breakdown or repair for any main or submain, water supply need not be stopped
In case of fire, large quantities of water can be drawn from all directions
Since water reaches every point from more than one route, the frictional losses and sizes of pipes can be reduced
<u>Disadvantages</u>
More number of valves and longer lengths of pipes are required
The overall cost is more
Design calculations are slightly tedious</td><td>3

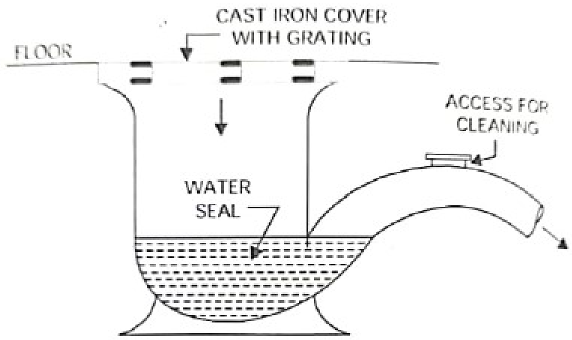
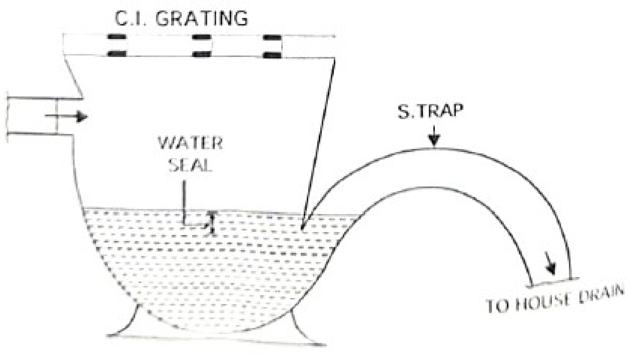
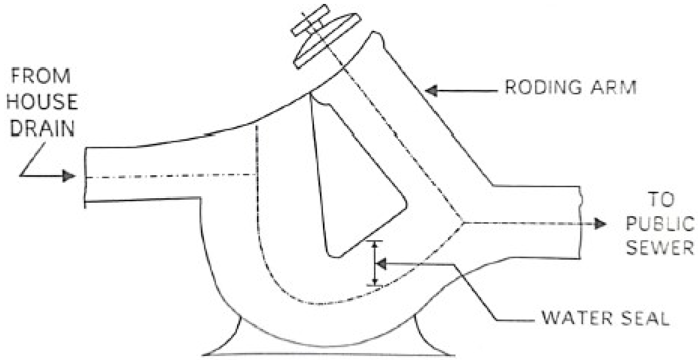
2

2</td><td>7</td></tr>
<tr><td>VI</td><td>It consists of a closed horizontal or vertical steel tankin which filter medium, gravel bed and under drainage system is provided
The diameter of tank varies between 1.5 to 3m and length between 3.5 to 8m
The water mixed with coagulant is directly fed to the filter. Filtered water is collected through under drainage system.
When filter is clogged, it is cleaned by back washing</td><td>3.5</td><td>7</td></tr>
</table>

	 PRESSURE FILTER	3.5	
VII	• It should react quickly in water to form flocs • It should be cheap • It should be easy to handle and store • Quantity should not deteriorate with time • It should give positively charged ions to attract negatively charged colloids • I should mix easily • It should react in the range of pH	Any 5 7	7
VIII	 Break point chlorination • The chlorine, when added to the water, performs the function of removing bacteria and then start accumulate up to certain point. This point is represented by point A on curve Q • If further dosage of chlorine is added in water, it is followed by a sudden decrease in residual chlorine content • The extra quantity addedof chlorine added after point A has been utilised for oxidisation of organic matter	2 5	7

	• If still further chlorine is addedin water, a point B is reached on curve Q. the point B on curve Q is known as break point as any chlorine that is added to the water beyond this point breaks through the water and appears as residual chlorine		
IX	• In drop manhole the branch sewer is joined to the man hole through a vertical pipe • The sewage trickles down the vertical pipe • A plug is provided at the point where branch sewer intersects the wall of manhole • The length of branch sewer between the vertical pipe and wall of manhole is known as the inspection arm and after opening the plug, it can be used for cleaning or inspection branch sewer  DROP MAN-HOLE	3 4	7
X	Water carriage system • Water is used as a medium for conveying sewage through the underground sewers. • In this system water closets provided with flushing arrangements Advantages 1. It is hygienic and decent system 2. No possibility of bad smell and fly nuisance 3. The sewage is properly treated and then disposed 4. No chance of ground water pollution Disadvantages 1. Its initial cost is high 2. It may require pumping of sewage at certain stages 3. It requires skilled supervision and maintenance	3 2 2	7

XI	1. To avoid pollution of receiving water bodies and thus preventing the health hazards 2. To create sanitary hygienic environment around the town 3. To protect the fish and other aquatic life 4. To avoid the sewage sickness of land on to which it is disposed 5. To derive the useful components after treatment of sewage 6. To prevent offensive odours and unsightly conditions of the water bodies used for swimming, boating etc.	Any 5 7	7
XII	1. Spreading on soil - the digested sludge may be disposed by spreading over farm land and ploughing after it has dried 2. Incineration - Involves total conversion of organic solids tooxidized end products. The sludge is burnt in either a flash type incinerator or multiple hearth type incinerator 3. Dumping – the digested sludge, grit and incinerator residue are dumped into an abandoned mine quarry 4. Ocean disposal- Dumping or controlledrelease of sewage sludge from a barge orother vessel into marine water 5. Heat drying – the sludge is dried artificially by applying heat. The dried sludge is disposed as a manure 6. Lagooning 7. Sanitary land fill	Any 5 7	7
XIII	• Considered as one of the alternate energy resources in rural areas • Unstable and putrescible organic solids are reduced into stable, acceptable form • The pathogenic organisms are destroyed during digestion • Ground water pollution is prevented • Helps in the maintenance of healthy environment in villages • Digested sludge acts as a good manure • House fly and mosquito breeding is eliminated	Any 5 7	7

XIV	 FLOOR TRAP  Gully trap  INTERCEPTING TRAP	2 2 3	7

Course:**Public Health Engineering**

BluePrint

MarkDistribution

Module	hr/ Module	Marks/Module $(h_i/\sum H_i)*123(\pm5\%)$	TypeofQuestions							
			PartA		Part B		PartC		Total	
			NoofQuestions	Marks	NoofQuestions	Marks	NoofQuestions	Marks	NoofQuestions	Marks
1	18	30	3	3	4	12	2	14	9	29
2	18	31	2	2	1	3	4	28	7	33
3	18	30	2	2	4	12	2	14	8	28
4	19	32	2	2	1	3	4	28	7	33
Total	73	123	9	9	10	30	12	84	31	123

BluePrint

CognitiveLevelMarkDistribution

CognitiveLevel	Marks	%ofMarks
Remembering	28	23
Understanding	95	77
Applying	0	0
Analysing		
Evaluating		
Creating		
Total	123	100

Question wise analysis

Course:**Public Health Engineering**

Q.No	ModuleOutcome	CognitiveLevel	Marks	Time
I.1	M 1.01	Remembering	1	2
I.2	M 1.02	Remembering	1	2
I.3	M 1.02	Remembering	1	2
I.4	M 2.04	Remembering	1	2
I.5	M 2.02	Remembering	1	2
I.6	M 3.02	Remembering	1	2
I.7	M 3.01	Remembering	1	2
I.8	M 4.04	Remembering	1	2
I.9	M 4.02	Remembering	1	2
II.1	M 1.02	Remembering	3	8
II.2	M 1.02	Understanding	3	8
II.3	M 1.01	Understanding	3	8
II.4	M 1.01	Understanding	3	8
II.5	M 2.02	Understanding	3	8
II.6	M 3.04	Remembering	3	8
II.7	M 3.01	Understanding	3	8
II.8	M 3.03	Remembering	3	8
II.9	M 3.02	Remembering	3	8
II.10	M 4.02	Understanding	3	8
III.	M 1.01	Understanding	7	16
IV.	M 1.04	Understanding	7	16
V.	M 2.05	Understanding	7	16
VI.	M 2.03	Understanding	7	16
VII.	M 2.02	Remembering	7	16
VIII.	M 2.02	Understanding	7	16
IX.	M 3.03	Understanding	7	16
X.	M 3.02	Understanding	7	16
XI.	M 4.01	Understanding	7	16
XII.	M 4.03	Understanding	7	16
XIII.	M 4.03	Understanding	7	16
XIV.	M 4.04	Understanding	7	16